

Milestone 2: Identifying Risk Factors for Postpartum Depression – Final White Paper

Jonathan Steen

MSDS, Bellevue University

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Instructor: Xu Ashton

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Executive Summary

This project examined whether non-outcome maternal, socioeconomic, family, obstetric, and neonatal variables could identify elevated postpartum depression screening risk. The analysis used 800 participants from Bangladesh. High risk was defined as an Edinburgh Postnatal Depression Scale (EPDS) score of 13 or higher. 349 participants (43.6%) met this threshold.

A regularized logistic regression model achieved a mean repeated cross-validated ROC-AUC of 0.774 and a holdout ROC-AUC of 0.760. A shallow decision tree achieved a holdout ROC-AUC of 0.731 and recall of 0.736. The results show moderate discrimination after screening items and closely related postpartum symptom proxies were removed. The appropriate use is exploratory risk stratification that directs attention toward validated screening and follow-up.

Recommendation: retain logistic regression as the analytical baseline, validate it prospectively in the intended population, and never substitute its output for EPDS screening, diagnosis, or clinical judgment.

Business Problem

Postpartum depression can affect maternal health, family functioning, and infant well-being. Healthcare teams need reliable ways to recognize risk early enough to support timely screening and follow-up. A risk model could help prioritize outreach when staff and behavioral health resources are limited. The business question is whether routinely available characteristics can identify elevated screening risk without using the symptoms that define the outcome.

Background and History

CDC reports that about one in eight U.S. women with a recent live birth reported symptoms of postpartum depression. This figure provides context and is not a benchmark for the Bangladesh sample. ACOG recommends depression and anxiety screening at the initial prenatal visit, later in pregnancy, and during postpartum care using standardized, validated instruments, with systems for assessment, treatment, monitoring, and follow-up.

Recent evidence describes postpartum depression as multifactorial. An umbrella review of 77 meta-analyses identified history of depression, poor social support, intimate partner violence, sleep disorders, and unintended pregnancy among the strongest reported risk factors. Most evidence was rated low or very low certainty, which limits causal interpretation. The current project tests predictive associations in one dataset and does not estimate causal effects.

Data Explanation

The primary source is Data for Postpartum Depression Prediction in Bangladesh, version 3, published through Mendeley Data under a CC BY 4.0 license. The release contains 800 respondents who had given birth within the prior 24 months and 70 columns covering demographics, income, family relationships, health history, pregnancy and delivery, newborn health, postpartum experiences, EPDS, PHQ-2, and PHQ-9.

Preparation and Data Quality

- The raw dataset contains 4,312 missing cells, including 4,089 among retained predictor fields before cleaning. Documented None responses and an unavailable or not-applicable older-child

level were assigned where supported. 206 ambiguous predictor blanks remained and were labeled Missing / not reported within each training fold.

- Clear spelling and capitalization variants were normalized. The invalid disease value 1.2 was recoded as missing. Categories occurring fewer than five times in a training fold were grouped with unseen levels as Other / infrequent.
- Categorical predictors were one-hot encoded. Numeric predictors were median-imputed and standardized for logistic regression comparability.
- No duplicate rows were found.
- One row labeled High had an EPDS score of 12. The documented numeric rule of 13 to 30 was applied, producing 349 high-risk cases.
- The final feature matrix contained 38 raw predictor columns before encoding.

Leakage Controls

All EPDS items, EPDS score and category, current PHQ-9 items, PHQ-9 score and category, and the row identifier were removed. Eight contemporaneous postpartum fields involving bonding, motherhood feelings, newborn worry, sleep or relaxation, irritability, and daily functioning were also removed because they could act as symptom proxies. PHQ-2 indicators for depression before and during pregnancy were retained as temporally prior mental health history.

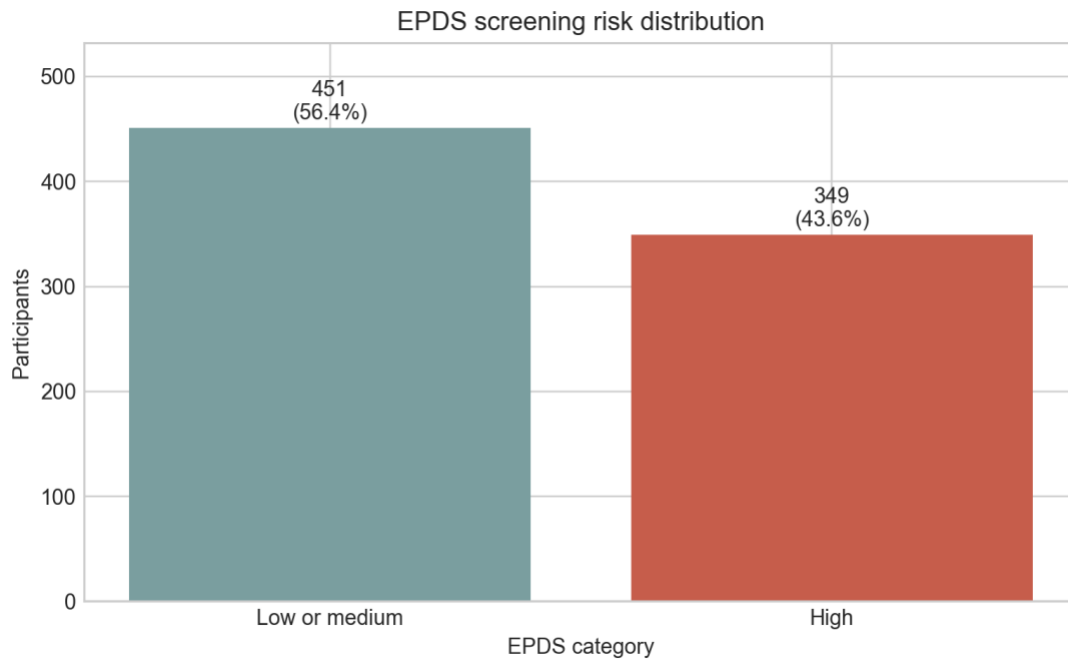


Figure 1. EPDS high-risk status based on the documented score threshold.

Methods

The analysis used a fixed random seed of 680 and a stratified 75/25 development and holdout split.

Model selection and stability were assessed on the 600-row development set with repeated stratified five-fold cross-validation using five repeats. The 200-row holdout set was used once for final comparison.

Two interpretable classifiers were compared. Logistic regression used L2 regularization and balanced class weights. The decision tree was limited to depth four with at least 25 observations per leaf and balanced class weights. Evaluation included ROC-AUC, average precision, recall, precision, F1, balanced accuracy, Brier score, and confusion-matrix counts. A 0.50 probability threshold was used for class predictions. Holdout ROC-AUC confidence intervals were estimated with 2,000 bootstrap samples.

Model	CV ROC-AUC	Holdout ROC-AUC	Recall	Precision
Logistic regression	0.774	0.760	0.598	0.650
Decision tree	0.729	0.731	0.736	0.552

Table 1. Leakage-controlled model performance.

Analysis

Descriptive Findings

The sample had a high EPDS screening-risk prevalence of 43.6 percent. Selected subgroup comparisons showed meaningful variation in high-risk prevalence, especially for depression screening before or during pregnancy and the ability to share feelings with trusted people. These rates are unadjusted and should be read as descriptive patterns.

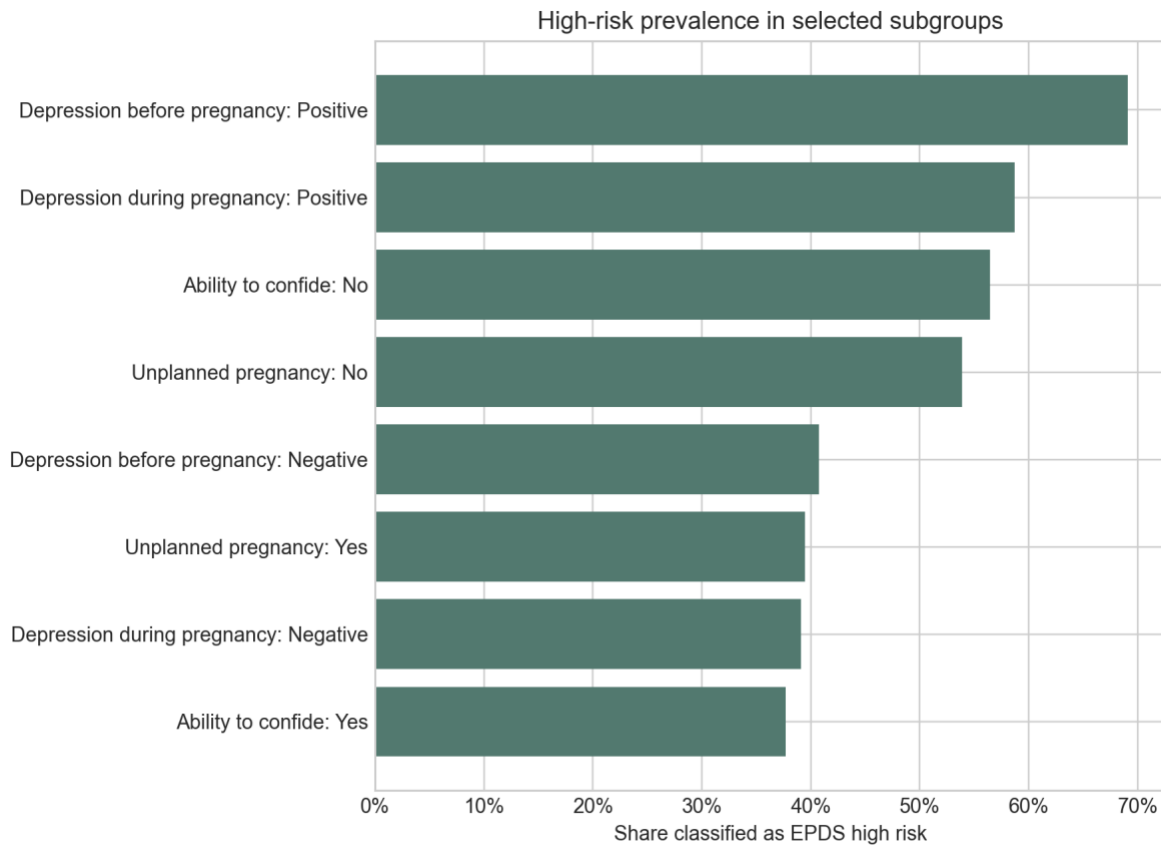


Figure 2. Unadjusted high-risk prevalence for selected subgroups with at least 20 observations.

Model Performance

Logistic regression provided the strongest cross-validated discrimination. Its holdout ROC-AUC was 0.760, with a bootstrap 95 percent interval of 0.690 to 0.826. At the 0.50 threshold, it identified 52 of 87 high-risk holdout participants and missed 35. The tree identified 64 high-risk participants and missed 23, producing higher recall and more false positives.

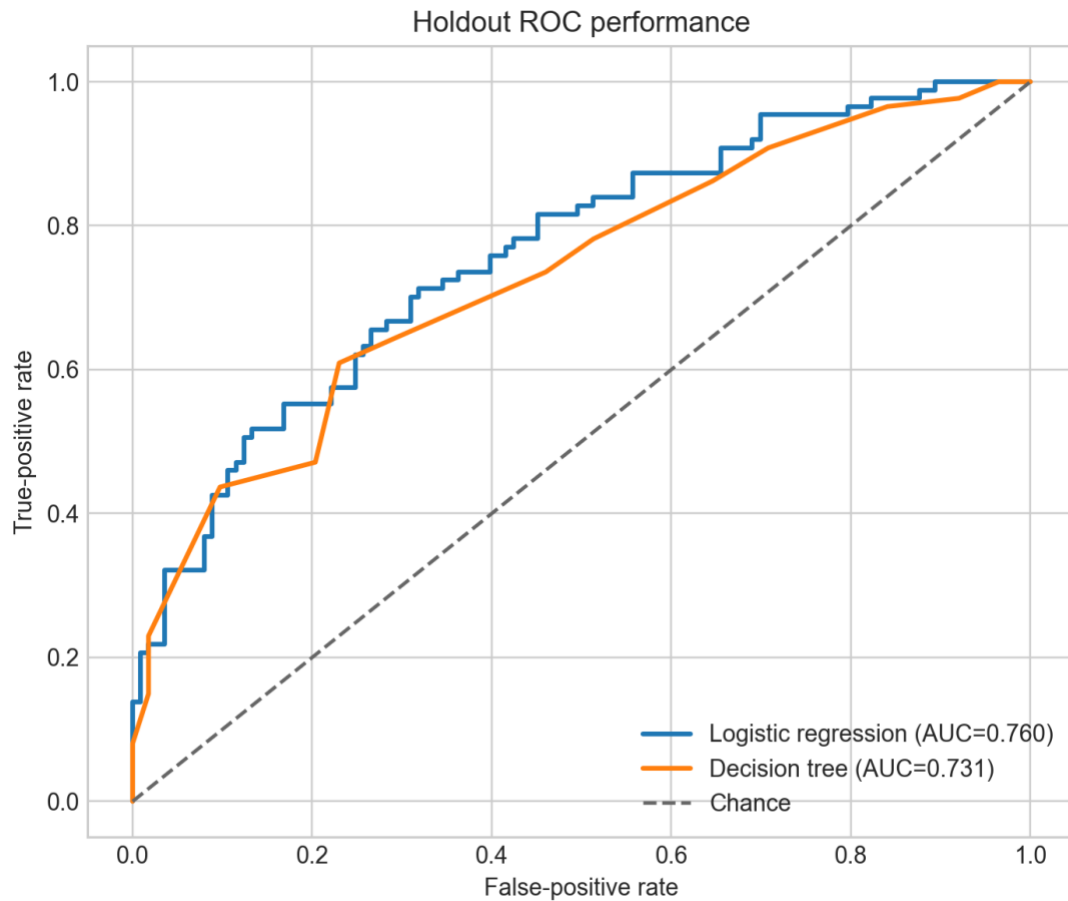


Figure 3. Holdout ROC curves for the two interpretable classifiers.

Interpretation

The fitted logistic model assigned substantial weight to prior and antenatal depression screening, pregnancy fear, major changes or losses, support, relationships, and selected obstetric or neonatal variables. Coefficients are conditional on the full encoded feature set and the chosen reference categories. Regularization shrinks estimates. They support interpretation of prediction, not causal claims or treatment decisions.

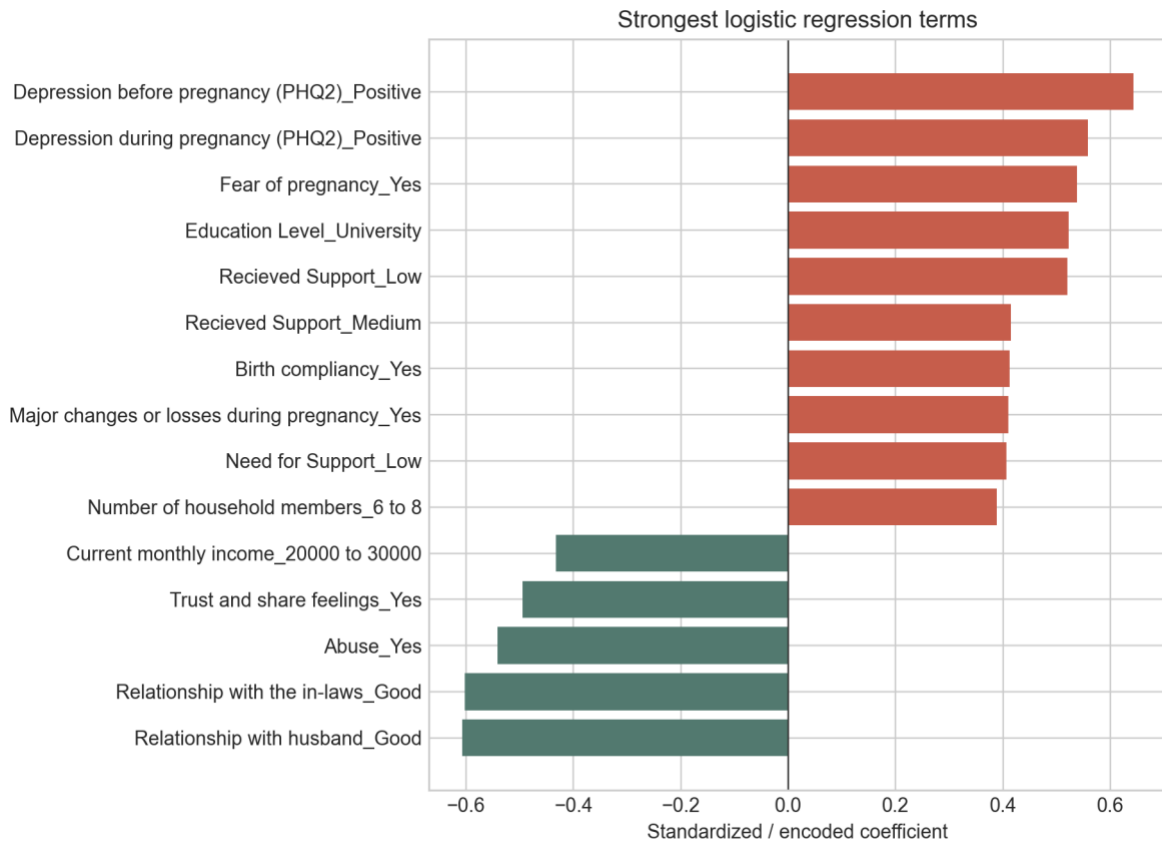


Figure 4. Largest absolute substantive coefficients in the regularized logistic regression model. Technical missing and rare-level terms are excluded from the display.

Conclusion

Postpartum depression screening risk can be predicted with moderate accuracy from non-outcome variables in this sample. Logistic regression was more stable across repeated validation, while the shallow tree recovered more high-risk cases on the holdout set. The modest performance is credible because direct screening fields and selected symptom proxies were excluded. The evidence supports further validation of an interpretable triage aid. It does not support clinical deployment or claims of diagnosis.

Assumptions

- An EPDS score of 13 or higher is an appropriate analytic threshold for elevated screening risk in this dataset.
- Prior and antenatal PHQ-2 indicators occur before the postpartum outcome and function as history rather than leakage.
- Documented None responses and unavailable fields are distinguishable from unresolved missing values; unresolved categorical blanks form an explicit Missing / not reported level.
- Rows represent independent participants, and the fixed holdout set was not used to select model settings.

Limitations

The cross-sectional, self-reported sample was recruited from facilities and communities in Bangladesh. Findings may not generalize to U.S. patients, other countries, or clinical workflows. Participants could be as far as 24 months postpartum, creating temporal heterogeneity. EPDS indicates screening risk and does not establish a clinical diagnosis. The sample size limits subgroup analysis, calibration assessment, and interaction testing. Missingness is substantial and its mechanism is undocumented. A single holdout split produces uncertainty, as reflected in the broad ROC-AUC intervals.

Challenges

The main technical challenge was distinguishing valid risk factors from fields that directly encode the outcome. A naive model could use EPDS or current PHQ-9 items and report excellent performance for the wrong reason. The dataset also contains spelling inconsistencies, multi-response fields stored as text, blanks with different meanings, rare categories, an invalid disease

value, and one mismatch between EPDS score and category. Column-specific missing semantics, fold-specific rare-level grouping, transparent exclusions, and a reproducible pipeline were necessary to preserve validity.

Future Uses and Additional Applications

- Prospective evaluation of prenatal and early-postpartum risk factors in a defined health system.
- Resource planning for behavioral health referrals, outreach, and follow-up capacity.
- Research on how social support and relationship measures interact with prior mental health history.
- Comparison of local calibration, subgroup performance, and threshold policies across care settings.

Recommendations

1. Use the logistic regression model as the analytical baseline because it had the strongest repeated validation and offers transparent coefficients.
2. Collect a prospective local dataset with clearly timed predictors, standardized missing-value definitions, and consent for model development.
3. Select a threshold with clinicians based on the cost of false negatives, referral capacity, and the role of validated screening.
4. Audit calibration and performance by age, residence, socioeconomic group, and other relevant populations before operational use.
5. Present risk estimates as prompts for screening and outreach. Do not present them as diagnoses or treatment recommendations.

Implementation Plan

Phase	Required work
1. Governance	Define intended use, accountable owner, privacy controls, consent, clinical escalation, and unacceptable uses.
2. Local data	Collect prospective data with predictor timestamps, missing-value codes, EPDS administration, and outcome follow-up.
3. Validation	Rebuild preprocessing, test discrimination and calibration, evaluate subgroups, and choose a threshold with clinical stakeholders.
4. Silent pilot	Run predictions without influencing care. Compare alerts with validated screening and measure workflow burden.
5. Limited rollout	Use the score only to prioritize outreach. Require human review and maintain standard screening for everyone.
6. Monitoring	Track drift, recall, false positives, calibration, subgroup disparities, referral completion, and patient complaints.

Ethical Assessment

The dataset includes mental health, abuse, income, relationship, and reproductive information. These variables require strict access controls and purpose limitation. A false negative could delay support, while a false positive could cause stigma, anxiety, or unnecessary intervention. The model may reproduce sampling and social inequities. Sensitive attributes should not be removed automatically, because they may be needed to measure disparity, but their operational use requires explicit justification.

Safe use requires informed governance, human review, validated screening, accessible follow-up, and a clear path for patients to ask questions or decline optional model-based outreach. Predictions should not be used for employment, insurance, custody, policing, benefit eligibility, or punitive decisions. Any self-harm response identified during standard screening requires an established clinical safety protocol rather than a model score.

Appendix A: Reproducibility, Data Dictionary, and Audience Q&A

Reproducibility

The final deliverables include an executed Jupyter notebook and a standalone Python analysis module.

The workflow uses a fixed random seed, documented category corrections, column-specific blank handling, fold-specific rare-level grouping, documented exclusions, pipeline-based encoding, repeated cross-validation, a fixed holdout set, and exported result tables. The raw files are retained with publisher-matching SHA-256 checksums.

Condensed Data Dictionary

Domain	Examples	Analytic role
Outcome	EPDS score	Binary high risk at score 13 or above
Demographic	Age, residence, education, marital status, occupation, income	Candidate predictors
Family and social	Family type, household size, relationships, support, abuse, trust	Candidate predictors
Health history	Disease, pregnancy loss, addiction, prior and antenatal PHQ-2	Candidate predictors
Pregnancy and birth	Pregnancy plan, checkups, fear, gestation, delivery, complications	Candidate predictors
Neonatal	Newborn age, sex, illness, breastfeeding	Candidate predictors
Excluded screens	EPDS items/results and current PHQ-9 items/results	Direct outcome leakage
Excluded proxies	Postpartum bonding, mood, worry, sleep, irritability, functioning	Closely related current symptoms

Audience Questions and Answers

1. Why was EPDS 13 selected as the high-risk threshold?

The data dictionary defines scores from 13 to 30 as high risk, so the analysis applied that rule consistently. The threshold creates an elevated screening-risk outcome. It does not establish a clinical diagnosis.

2. How much would performance change if postpartum symptom proxies were included?

The final analysis did not estimate that change because the excluded postpartum fields closely reflect current symptoms or functioning. Adding them would answer an easier and less useful question and could inflate apparent performance. A separate sensitivity analysis could quantify the inflation, but it should be labeled as a leakage audit.

3. Why retain PHQ-2 history while excluding current PHQ-9 items?

The retained PHQ-2 indicators describe depression screening before or during pregnancy, which places them before the postpartum outcome. Current PHQ-9 items measure contemporaneous depressive symptoms and would overlap directly with the outcome being predicted.

4. Which model should be used if missing a high-risk patient is more costly than a false alert?

At the fixed 0.50 threshold, the decision tree had higher recall (0.736) than logistic regression (0.598) and identified 64 of 87 high-risk holdout participants. It also generated 52 false positives compared with 28 for logistic regression. A safer next step is to retain the more stable logistic

model and lower its threshold during prospective validation, using referral capacity and the cost of missed cases to select the operating point.

5. How should the extensive missing data be interpreted?

The retained predictors contained 4,089 raw missing cells. Documented None responses and structural non-applicability were mapped explicitly, leaving 206 ambiguous blanks as Missing / not reported. Missingness may carry information, but its mechanism is undocumented and could introduce bias.

6. Can findings from Bangladesh be applied to U.S. patients?

Direct application to U.S. patients is not supported. The sample reflects a specific Bangladesh population, recruitment process, care environment, and social context. U.S. use would require local data collection, model rebuilding, external validation, calibration assessment, and subgroup evaluation.

7. Does a high model score mean that a patient has postpartum depression?

A high score means the model estimates a greater probability of an EPDS score of 13 or higher. It does not confirm postpartum depression. A validated screening instrument, clinical assessment, and appropriate follow-up remain necessary.

8. Which predictors were most influential, and are their relationships causal?

The strongest model terms included depression screening before and during pregnancy, pregnancy fear, support and relationship measures, major changes or losses, and selected obstetric or neonatal factors. These are conditional predictive associations within a regularized model. They do not establish that changing a factor would change the outcome.

9. How would fairness and subgroup performance be evaluated before implementation?

Evaluation should report discrimination, calibration, recall, precision, false-negative rates, and false-positive rates by predefined groups. Confidence intervals and group sample sizes should accompany every comparison. Thresholds should be reviewed with affected communities and clinical stakeholders, followed by ongoing drift and disparity monitoring.

10. What evidence would be required before using the model in clinical workflow?

Required evidence includes prospective local validation, reliable calibration, a clinically selected threshold, acceptable subgroup performance, a silent workflow pilot, privacy and consent controls, clear accountability, and demonstrated access to screening and follow-up. The model should remain a triage aid and should never replace standard screening or clinical judgment.

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